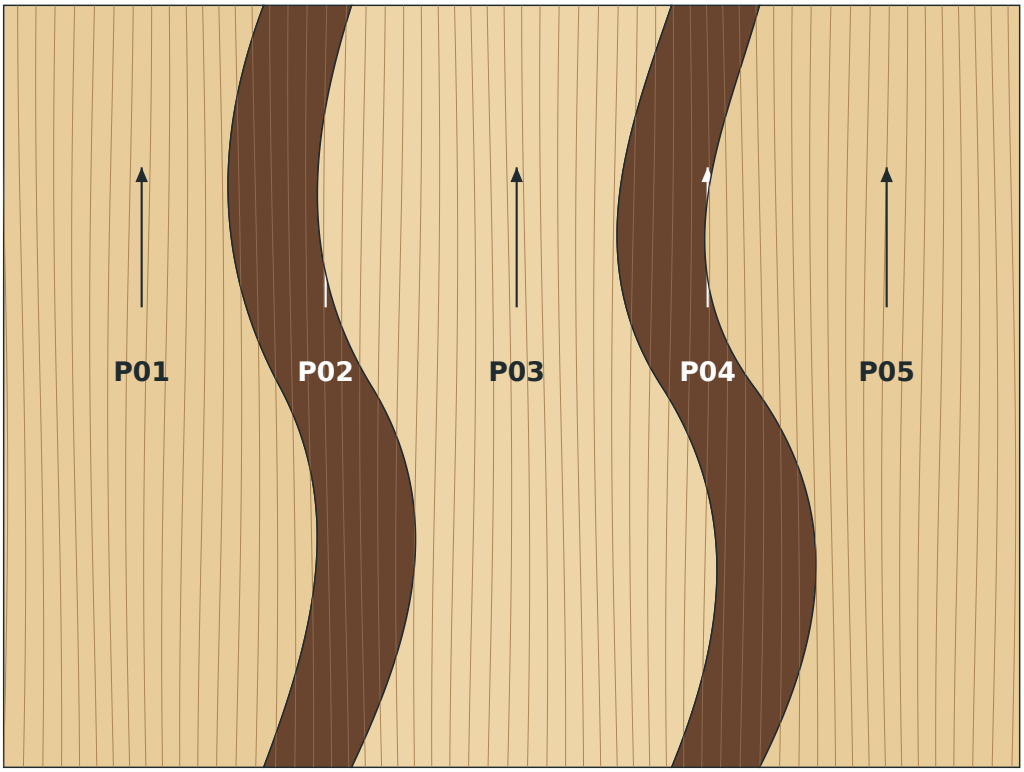
Your practice project

Build an editable CNC workflow for a five-piece maple and walnut cutting board. This pack supplies the client-style inputs; you create the native VCarve jobs, machining previews and evidence.



Start with these four actions

- Read the client brief in folder 01 and the drawings in folder 02.
- Open folder 08 and download the genuine VCarve starter projects.
- Import the warm-up DXF, then the two separate joint-coupon files.
- Create the five board jobs, save editable CRV files and record your own previews.

What is supplied: original DXF/SVG geometry, reference images, dimensions, stock layouts, a joint test, editable notes, templates and an optional generator.

What you will create: native CRV files and CAM. Actual machining, measured fit, machine-specific code and the recorded walkthrough remain practical milestones. No custom CRV or test-cut evidence is fabricated.

Read the client handoff

The fictional client is River Practice Workshop. The job description gives the size, five adjoining pieces and maple/walnut materials. This exercise fills the remaining gaps with explicit assumptions; it is not the actual River No. 1 design.

Requirement	Practice definition
Finished board	508 x 381 x 25.4 mm
Pretrim glue-up	514 x 387 x 28.0 mm
Material sequence	P01 maple / P02 walnut / P03 maple / P04 walnut / P05 maple
Grain	All five pieces parallel to +Y, along 381 mm
Joint construction	Four smooth, full-thickness, vertical mating edges
Nominal geometry	Zero CAD gap; CAM fit allowance tested separately
Shop for simulation	Virtual 3-axis router, at least 600 x 600 mm XY
Controller and post	Unselected; code release requires the actual setup

What the simulated shop does

The shop prepares flat blanks, checks moisture, fixtures and machines parts, dry assembles, glues with cauls and a common end datum, then flattens and trims. Tabs stay in sacrificial end waste. No handles, juice groove, inlay or hidden joinery is included.

How acceptance is defined

The practice dry-fit target is hand assembly without force and a provisional 0.10 mm feeler check. Proposed finished tolerances are +/-0.5 mm in plan and +/-0.3 mm in thickness. Repeat three accepted coupon pairs before claiming repeatability. The full criteria are in the client brief.

Before the real job: replace each practice assumption with the client's approved value. The assumption register in folder 01 distinguishes supplied facts, invented exercise values and open release decisions.

First native file and small test

1 Inspect an official CRV

Download R01 from the resource page in folder 08. Open Welcome Sign.crv with File > Open. Save a copy, inspect job settings and existing toolpaths, make one small edit, recalculate and preview. R02 then shows the DXF-to-profile route closest to the board task.

2 Make the warm-up plate

Create a 140 x 100 x 12 mm single-sided job. Import Warmup_Plate_Stock_mm.dxf. The part is 100 x 60 mm with two 10 mm holes. Its lower-left is 20,20; hole centers are 40,50 and 100,50. Pocket holes before the outside cut, preview and save your first CRV.

3 Build the two coupon jobs

Use the separate stock files from folder 04. The assembled pair measures 160 x 120 mm and reuses the middle of the actual J12 curve at full scale. Both blanks are 28 mm thick.

Part	Material	Stock X x Y
A_Maple	Hard maple	140 x 160 mm
B_Walnut	Walnut	145 x 160 mm

Set XY zero at the lower-left stock corner and Z zero at the stock top. Preserve the supplied placement. If import shifts the drawing, align the entire imported set using REF_STOCK before hiding it.

Select only CUT_COUPON_A_MAPLE or CUT_COUPON_B_WALNUT. The outline is the part edge, so use an outside profile with cutter compensation in CAM. Do not add a second tool-radius offset to the DXF.

Checkpoint: one correct native job for each material, one closed machining outline per part, correct stock dimensions and no toolpath around the REF_STOCK rectangle.

Prepare all five board parts

Use one native job per part for the first exercise. This makes material, zero position and stock changes easy to inspect. Multi-sheet organization can follow after you understand the basic process.

ID	Wood	Stock X x Y x Z	Machining layer
P01	Hard maple	200 x 430 x 28	CUT_P01_MAPLE
P02	Walnut	135 x 430 x 28	CUT_P02_WALNUT
P03	Hard maple	240 x 430 x 28	CUT_P03_MAPLE
P04	Walnut	140 x 430 x 28	CUT_P04_WALNUT
P05	Hard maple	205 x 430 x 28	CUT_P05_MAPLE

- Import the matching Stock_Layout_mm.dxf in millimetres at 1:1. Check REF_STOCK against the job boundary, then hide reference layers.
- Profile only the CUT_Pxx_MATERIAL contour. Use the nominal baseline until the coupon process is physically accepted.
- Review retention, entry, exit and any end-waste tabs. No tab or entry mark should compromise a mating surface.
- Preview and save RP_P01_Maple_A.crv through RP_P05_Maple_A.crv. Export a VCarve job sheet and capture your own previews.
- Dry assemble the physical full board only after cutting. Flatten, trim and soften external edges after glue-up under the shop's agreed process.

Finished versus pretrim: master finished geometry is 508 x 381 mm. Individual machining files contain 3 mm external trim waste; the untrimmed assembly is 514 x 387 mm. Do not scale the stock-layout parts to the finished size.

Reference assembly: touching outlines in a master drawing are for review. Cut each material-specific component from its own blank with its own compensated profile.

Set up a controlled fit test

Use these virtual values to learn the interface. They have not been validated for a real cutter, machine, fixture or piece of wood. The detailed lab in folder 05 explains the checks before any physical cut.

Field	Simulation value
Tool	T01 flat end, 6.35 mm, two flutes, adequate cutting length
Speed / feed / plunge	18000 rpm / 1800 mm/min / 400 mm/min
Stock / start / cut depth	28.00 / 0.00 / 28.20 mm, with sacrificial spoilboard
Maximum roughing stepdown	2.00 mm; inspect actual pass list
Baseline fit / last-pass allowance	0.000 mm / 0.15 mm radial
Safe clearance	15 mm above stock for simulation; verify actual fixtures
Example tab settings	12 mm length, 2.2 mm toolpath thickness; 2.0 mm stock remains

Compare three B variants

Keep A nominal. Save B00 at zero allowance, B025 at -0.025 mm and B050 at -0.050 mm. A negative outside-profile allowance makes B smaller. The whole B outline changes in this simple test; keep its original vectors.

Cut only after operator review. Check top, middle and bottom gap, edge squareness, movement and tear-out. Log actual results in Test_Results.csv. Then repeat three accepted pairs and finally test the full board.

Keep three ideas separate: cutter radius compensation, temporary finishing stock and final fit allowance. A mathematically matching curve does not establish a physically matching joint.

The last pass can engage the full 28 mm edge height; the small roughing stepdown does not prove this is safe. Use the cutter maker's guidance and the actual operator's test plan.

Keep the handoff editable

Your first automation should reduce repeated setup while leaving geometry and toolpaths understandable to the client. The five-piece project does not justify a large custom automation system by itself.

Work	Recommended route
Shared curves and dependent parts	External parameter master or a clearly managed native vector master
DXF/SVG export and stock schedule	Included standard-library Python generator
Tool definitions, datums and strategy	VCarve plus operator review
Reuse of accepted operations	VCarve toolpath templates
Tabs, entries, preview and revisions	Inspect in VCarve for each revision
Job sheets and documentation	VCarve job sheet plus the supplied setup template
Future repetitive tasks	Consider a Lua Gadget only after the manual process works

Try one concrete client revision

Change width from 508 to 520 mm, keeping height 381 mm. Copy parameters.json to a new revision, update width and revision, and run the generator into a fresh output folder. Confirm the new master measures 520 x 381 mm.

Import the new vectors into copies of the CRV jobs. Reselect or relink geometry, recalculate, review stock and retention, and save new setup sheets. The script does not create native CRV, toolpaths or machine code.

One master per revision: direct VCarve edits do not update the JSON. Decide which source is authoritative before the next change; do not regenerate over a client's manual edits.

Download real VCarve examples

Open the HTML index in folder 08 for direct ZIP downloads and the matching videos. All three archives were downloaded and checked for native files on 8 November 2026. They were not opened in VCarve here.

R01 Create a Simple Sign

Native file: Welcome Sign.crv
Learn the job setup, drawing tools, machining preview and native-file structure.

Official project and video | Download original ZIP

R02 Import Vectors and Create Profile Toolpaths

Native file: Garden Panel - Toolpaths.crv
Closest starter to this board: inspect a DXF import, sheets and a profile cut.

Official project and video | Download original ZIP

R03 Create a V-carved Sign

Native file: Backyard BBQ Sign.crv
Practise editing vectors, recalculating and sending a revised preview. V-carving itself is optional here.

Official project and video | Download original ZIP

Official project pages list V12 or later. Exact creation revisions of the binaries were not independently established. If a direct link changes, use its project page. The optional downloader extracts learning inputs and skips machine code.

Use these references while you work

R04 Quick Start Guide

R05 2D Profile Toolpath

R06 Toolpath Templates

R07 Create Job Sheet

R08 Gadgets and Lua

Original versus reference work: this pack's board is your independent practice material. Official Vectric project files remain with Vectric and are learning examples, not samples of work you personally designed.

Turn the exercise into evidence

Your ArtCAM experience is a useful starting point for stock, cutter and toolpath concepts. This exercise helps test how quickly you can organize an editable VCarve workflow. It does not establish hands-on hardwood CNC experience by itself.

Milestone	Evidence required
1 Digital familiarization	Open a genuine CRV, make an edit, recalculate and save a preview
2 Coupon CAM	Two material-specific native jobs, allowance variants and setup notes
3 Five-part CAM	Five editable jobs, stock schedule, job sheets and revision example
4 Physical validation	Actual coupon measurements, photos and repeated cuts
5 Real production release	Accepted full board and verified machine/controller/post/fixture

What to send a prospective client

Use the editable case-study template in folder 09. Show your own VCarve screenshots and native jobs, explain the shared-boundary and coupon approach, and list physical cutting as pending unless performed. A 3 to 5 minute walkthrough plan is included.

Before a firm quote

Confirm approved geometry, thickness, stock, grain, preparation, assembly, acceptance tolerances, VCarve version, machine/controller, cutter details, workholding, who runs tests, feedback timing and the prototype revision limit.

Role split: you can lead geometry, file organization and automation. Your collaborator should own or review woodworking decisions. If neither person has CNC machining experience, add an experienced operator before promising production-ready files.

Scope discipline: time the exercise first. Quote a small design-and-coupon milestone, then the full board and machine-specific release. Do not turn tutorial files into claimed personal portfolio work.